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Abstract

Plastic waste management remains a critical environmental challenge due to inefficient tracking, poor segregation, and low recycling rates. By using a Digital Product Passport (DPP) system integrated with QR code-based tracking to improve transparency and accountability across the plastic product lifecycle. Each plastic item is assigned a unique QR code that stores essential information such as material type, manufacturing details, usage guidelines, and end-of-life disposal instructions. Stakeholders, including consumers, manufacturers, and waste management authorities, can access this data in real time using a simple scan.

The proposed system enhances traceability from production to recycling, enabling better monitoring of waste flow and identification of inefficiencies in the disposal chain. It promotes responsible consumer behavior, improves segregation at source, and supports recycling facilities with accurate material data, thereby increasing resource recovery efficiency. Integration with secure digital databases or blockchain ensures data integrity and reliability. By aligning with circular economy principles, the system encourages sustainable production practices and reduces environmental pollution. Overall, this QR-based DPP framework offers a scalable and technology-driven solution for modernizing plastic waste management systems and fostering environmental sustainability.

INTRODUCTION

1.1 Overview

The global economic system has historically operated on a linear "take–make–dispose" model in which raw materials are extracted, transformed into products, and eventually discarded as waste after consumption [21]. This model has led to significant environmental degradation, resource depletion, and unsustainable production practices. In response to these challenges, the concept of a Circular Economy has emerged as a sustainable alternative that emphasizes reuse, recycling, remanufacturing, and extending product lifecycles [21]. However, implementing a circular economy requires accurate lifecycle tracking, reliable data sharing, and trust among multiple stakeholders, which remain major challenges in existing systems. Technologies such as blockchain and digital product passports offer promising solutions by enabling secure, transparent, and immutable record-keeping across supply chains [6, 7]. In this project, a blockchain-enabled system is proposed where Digital Product Passports store lifecycle information, while financial transactions are used as the primary mechanism for validating real-world actions [5]. Unlike traditional systems, QR codes are used only for authentication and retrieval of product identity, while lifecycle updates are triggered through verified payment events recorded on the blockchain [5].

1.2 Motivation

The motivation for this project arises from the limitations of current sustainability and waste management systems, which often lack transparency, accountability, and reliable verification mechanisms. Many organizations claim adherence to circular practices, but there is no robust system to validate whether recycling or reuse actually occurs [4]. Existing blockchain-based approaches still rely heavily on QR code scanning, which depends on user participation and can be easily manipulated or bypassed [3, 4]. Furthermore, fragmented data systems and lack of trust between stakeholders lead to inefficiencies and misinformation [8]. A key realization is that financial transactions, unlike scans or manual entries, are inherently difficult to falsify and provide stronger proof of real-world activity. By linking lifecycle events such as recycling and material transfer to payment transactions, the system ensures that each recorded action corresponds to a tangible economic exchange, thereby improving trust, accountability, and data reliability [5].

1.3 Objective

The primary objective of this project is to design and develop a comprehensive blockchain-enabled framework that supports the implementation of a circular economy through secure, transparent, and verifiable lifecycle tracking of products. The system aims to establish a robust mechanism for creating and managing Digital Product Passports, which serve as unique digital identities containing detailed information about a product's origin, composition, ownership, and lifecycle events [5]. By maintaining this information on a decentralized ledger, the system ensures data integrity, immutability, and accessibility for all authorized stakeholders across the supply chain [7].

A key objective of the proposed system is to move beyond traditional scan-based validation mechanisms and introduce a more reliable method of verifying real-world actions through payment-linked events. In this approach, financial transactions such as incentives, settlements, or transfers act as proof of actual physical activities, including product return, recycling, or ownership transfer. These transactions are recorded on the blockchain and validated using smart contracts, which automatically update the lifecycle state of the product [3, 7]. This ensures that every recorded event is backed by tangible economic activity, thereby reducing the possibility of fraudulent or inaccurate data entries and significantly improving trust among stakeholders.

Another important objective is to enable end-to-end lifecycle visibility, allowing all participants in the supply chain, including manufacturers, distributors, consumers, collection centers, and recyclers, to access accurate and real-time information about a product's status [5]. This transparency facilitates better decision-making, improves coordination among stakeholders, and supports regulatory compliance by providing verifiable and auditable records. The system is also designed to incentivize sustainable behavior by integrating reward mechanisms that provide financial or token-based benefits to consumers and organizations that participate in proper disposal and recycling practices.

1.4 Scope

The scope of this project encompasses the design and conceptual development of a blockchain-enabled circular economy framework specifically applied to plastic bottle lifecycle management, while remaining flexible enough to extend to other materials and industries [5]. The system covers the entire product journey, beginning from manufacturing and distribution to consumption, collection, and recycling, ensuring that each stage is digitally recorded and verifiable. A key aspect within the scope is the implementation of Digital Product Passports that uniquely identify each product and

store its lifecycle data in a structured and secure manner. QR codes are incorporated strictly as a lightweight authentication mechanism to access product information, ensuring ease of use without influencing lifecycle state changes [5]. The scope also includes the integration of a payment-linked validation layer, where real-world actions such as product returns and recycling are verified through financial transactions, thereby ensuring authenticity and eliminating dependence on manual or scan-based confirmations.

In addition to lifecycle tracking, the scope extends to enabling stakeholder interaction within a decentralized environment, including manufacturers, distributors, consumers, collection centers, and recyclers. The system is designed to support token-based incentive mechanisms that encourage sustainable user behavior, such as proper disposal and participation in recycling programs. It also includes provisions for generating transparent and auditable reports that can be used for regulatory compliance, environmental impact assessment, and ESG reporting. While the current implementation is conceptual, the architecture is designed with future extensibility in mind, allowing integration with IoT devices for real-time data capture, enterprise systems for operational scalability, and advanced analytics tools for decision-making support [5]. This ensures that the project not only addresses current challenges but also adapts to evolving technological and regulatory landscapes.

1.5 Existing System

Existing systems that aim to implement circular economy principles using digital technologies primarily rely on basic blockchain architectures combined with QR-based tracking mechanisms [8, 10, 11]. These systems focus on recording lifecycle events such as production, distribution, and recycling on a decentralized ledger, thereby ensuring data immutability and traceability [6]. However, the validation of these events is typically dependent on manual inputs or QR code scans performed by users or stakeholders at different stages of the supply chain [4]. While this approach provides a level of transparency, it fails to guarantee that the recorded events accurately reflect real-world actions, as scans can be skipped, duplicated, or manipulated [3]. Additionally, many existing implementations are built on public blockchain platforms, which often face challenges related to scalability, high transaction costs, and limited privacy controls, making them less suitable for large-scale industrial adoption [6, 20].

Another significant limitation of current systems is their inability to seamlessly integrate with existing enterprise infrastructure and financial ecosystems. Most solutions treat lifecycle tracking and financial transactions as separate processes, resulting in fragmented workflows and reduced efficiency [8]. Furthermore, the lack of standardized

frameworks for Digital Product Passports leads to inconsistencies in data representation and interoperability issues across different platforms [12]. While some advanced systems attempt to incorporate smart contracts and automation, they still rely on weak verification signals such as user input rather than strong proof mechanisms [3]. As a result, these systems struggle to establish trust among stakeholders and fail to provide reliable evidence for sustainability claims, highlighting the need for a more robust and integrated approach.

1.6 Proposed System

The proposed Digital Product Passport (DPP) system integrated with QR-based tracking for plastic waste management aims to create a transparent, traceable, and efficient ecosystem for handling plastic materials throughout their lifecycle [5]. By assigning a unique QR code to each plastic product, the system will enable seamless access to critical information such as material composition, manufacturing details, usage guidelines, and end-of-life disposal instructions [5]. This will significantly improve awareness among consumers and stakeholders, encouraging responsible usage and proper segregation of waste at the source.

One of the primary achievements of the system will be enhanced traceability. Every stage of a plastic product's journey—from production to recycling—can be digitally recorded and monitored. This will help identify inefficiencies, leakage points, and improper disposal practices, thereby enabling targeted interventions. It will also support regulatory authorities in enforcing environmental compliance more effectively [1].

The system will also promote circular economy practices by facilitating better recycling rates and reducing contamination in waste streams [22]. Waste collectors and recycling facilities will benefit from accurate data, allowing for more efficient sorting and processing. This, in turn, will increase the recovery of valuable materials and reduce dependence on virgin plastic production.

Additionally, the proposed system will foster accountability across the supply chain. Manufacturers will be encouraged to adopt sustainable materials and eco-friendly designs due to increased visibility of product lifecycle impacts. The integration of digital tracking technologies such as cloud databases or blockchain will further ensure data reliability and trust [7, 5].

Overall, the system will achieve a more organized, data-driven, and environmentally responsible plastic waste management framework, ultimately contributing to reduced pollution and a more sustainable future.

PROBLEM STATEMENT

2.1 Problem Statement

The current global production and consumption model is predominantly linear in nature, following a "take–make–dispose" approach that leads to excessive waste generation, inefficient resource utilization, and significant environmental degradation [21]. Although the concept of a circular economy has been introduced to address these challenges by promoting reuse, recycling, and sustainable resource management, its practical implementation remains limited due to the absence of reliable systems for tracking product lifecycles [4, 20]. Existing mechanisms fail to provide accurate, transparent, and verifiable information about whether materials are genuinely being reused or recycled, resulting in a lack of accountability across the supply chain [3, 4]. This creates a major barrier to achieving true circularity, as stakeholders cannot confidently verify sustainability claims or ensure compliance with environmental standards.

A major limitation of current digital solutions lies in their reliance on weak verification methods such as QR code scanning, manual data entry, or self-reported updates [4]. While these methods enable basic tracking, they do not guarantee that recorded events correspond to actual physical processes, as they can be easily manipulated, skipped, or falsely recorded [3]. Even blockchain-based systems, despite offering immutability and transparency, often depend on these unreliable inputs, thereby compromising the integrity of the entire system [6]. Additionally, the lack of integration between lifecycle tracking and financial systems further reduces the effectiveness of these solutions, as there is no strong linkage between recorded data and real-world economic activity. This results in fragmented systems that fail to establish trust among stakeholders and limit the adoption of circular economy practices [8].

Therefore, there is a critical need for a robust and trustworthy system that can ensure accurate lifecycle tracking and verifiable validation of real-world actions. The problem addressed in this project is the development of a blockchain-based framework that replaces traditional scan-based verification with payment-linked validation, where financial transactions serve as proof of actual activities such as product return, collection, and recycling. By leveraging blockchain technology, smart contracts, and Digital Product Passports, the system aims to create a secure, transparent, and tamper-proof environment that enhances trust, reduces fraud, and enables effective implementation of circular economy principles [5].

2.2 Motivation

The motivation for this project arises from the growing environmental and economic challenges associated with inefficient waste management and the lack of accountability in current circular economy implementations. Despite increasing awareness about sustainability, many industries continue to struggle with verifying whether materials are genuinely recycled or reused, leading to widespread greenwashing and unreliable reporting [4]. Existing systems often rely on manual processes, fragmented databases, or QR-based tracking, which do not provide strong guarantees of real-world actions [4, 8]. This lack of reliable verification not only reduces trust among stakeholders but also discourages active participation from consumers and organizations, as there are limited incentives and no transparent mechanism to validate their contributions to sustainability.

Another key motivating factor is the realization that financial transactions inherently carry stronger credibility compared to traditional tracking methods. While QR scans and manual entries can be easily manipulated or skipped, monetary exchanges represent tangible actions that are difficult to falsify and are already governed by secure and regulated systems. By linking lifecycle events such as product return and recycling to payment-based triggers, it becomes possible to create a more trustworthy and automated verification system [5]. This approach not only enhances data integrity but also aligns economic incentives with sustainable behavior, encouraging stakeholders to actively participate in circular practices. Furthermore, advancements in blockchain technology provide an ideal foundation for integrating such mechanisms, offering immutability, transparency, and decentralized trust [6, 7].

The motivation also extends to the need for a scalable and future-ready solution that can integrate with emerging technologies and regulatory frameworks. With increasing emphasis on ESG compliance and sustainable supply chains, organizations require systems that can provide accurate, auditable, and real-time data [14, 18]. The proposed system addresses this need by combining blockchain, Digital Product Passports, and payment-linked validation into a unified framework that supports transparency, efficiency, and accountability [5]. This not only bridges the gap between theoretical circular economy concepts and practical implementation but also creates opportunities for innovation in waste management, supply chain optimization, and sustainable development.

2.3 Objectives

The primary objective of this project is to develop a comprehensive and reliable system that enables effective implementation of circular economy principles through secure lifecycle tracking and verifiable validation of product-related activities. The system aims to establish a decentralized framework where Digital Product Passports are used to uniquely identify products and store their lifecycle information in an immutable and transparent manner [5]. By leveraging blockchain technology, the system ensures that all recorded data is tamper-proof, accessible to authorized stakeholders, and capable of supporting end-to-end traceability across the supply chain [7, 5].

A central objective of the proposed approach is to replace traditional scan-based or manual verification mechanisms with payment-linked validation, thereby ensuring that lifecycle events are directly tied to real-world economic activity. In this system, financial transactions such as incentives, settlements, or transfers act as proof of actions like product return, collection, and recycling. Smart contracts are used to automatically validate these transactions and update the corresponding product states on the blockchain, eliminating the need for manual intervention and reducing the risk of errors or fraud [3]. This objective not only improves the accuracy and reliability of lifecycle tracking but also introduces a novel method of integrating financial systems with blockchain-based sustainability solutions.

Another important objective is to design a scalable, flexible, and user-friendly architecture that can be adapted across different industries and use cases [8]. The system aims to support seamless integration with existing enterprise platforms, digital payment systems, and emerging technologies such as IoT and digital twins [27, 9], ensuring long-term applicability and evolution. Additionally, the project seeks to promote sustainable behavior by incorporating incentive mechanisms that reward users for participating in recycling and responsible consumption practices [25, 22].

DETAILED SURVEY

The detailed survey presents an analysis of existing research works and technologies related to blockchain-enabled circular economy systems [5, 3, 4, 1, 22, 25]. Various approaches involving blockchain technology, Digital Product Passports (DPPs), smart contracts, decentralized identity systems, and recycling incentive mechanisms have been studied to understand their role in improving transparency, traceability, and sustainability in product lifecycle management.

This survey helps identify the strengths and limitations of current systems and highlights the research gap addressed by the proposed project. The analysis mainly focuses on lifecycle tracking methods, validation mechanisms, stakeholder interaction, and integration of blockchain with sustainable waste management practices [8, 19, 20, 17].

SURVEY SUMMARY TABLE

Each entry below presents one surveyed paper as a labelled block, reading top-to-bottom on a standard portrait page.

Sl. No.	1
Title	Integrating Blockchain with Digital Product Passports for Managing Reverse Supply Chain — H. Xia et al. (2025) [5]
Problem Addressed	Inefficient reverse supply chain management caused by poor product traceability, fragmented data sharing, and limited lifecycle information, proposing blockchain-enabled digital product passports to improve transparency and recycling efficiency.
Author's Approach/Method	An extensive literature survey over 3 reputed citation databases—Google Scholar, Web of Science and Scopus. High quality journal papers (impact factor >3) were used. Keywords: "reverse supply chain" + "blockchain" + "digital product passport". Selection refined to 100 after filtering; further screening resulted in 34 papers.
Result	Blockchain-enabled DPPs improve lifecycle traceability, information sharing, and reverse logistics efficiency, leading to better recycling, remanufacturing decisions, and overall reverse supply chain performance [5].

Sl. No.	2
Title	Leveraging Blockchain and Smart Contract Technologies to Overcome Circular Economy Implementation Challenges — N. M. Kumar and S. S. Chopra (2022) [3]
Problem Addressed	Operational and coordination challenges in circular economy implementation, such as lack of trust, transparency, and automation in supply chains.
Author's Approach/Method	Analyzes implementation challenges, proposes a smart contract framework, and evaluates its feasibility through theoretical modelling and literature analysis.
Result	Proposes a Circular Economy Blockchain (CEB) architecture showing that blockchain and smart contracts can enhance transparency, trust, traceability, and stakeholder collaboration in circular supply chains [3].

Sl. No.	3
Title	Blockchain for the Circular Economy: Analysis of the Research-Practice Gap — A. Böckel, A. K. Nuzum, and I. Weissbrod (2021) [4]
Problem Addressed	Operational and coordination challenges in circular economy implementation; lack of trust, transparency, and automation.
Author's Approach/Method	Collected 57 documents from academic papers, industry reports, and online sources. Conducted a research-practice gap analysis categorising data to identify patterns and differences between academic research and real-world applications.
Result	Highlights blockchain's potential to address sustainability challenges; identifies lack of clear terminology and emphasises trust and verification as significant yet challenging advantages [4].

Sl. No.	4
Title	Applying the Circular Economy Framework to Blockchain Agricultural Production (2024) [1]
Problem Addressed	Inefficiencies, lack of transparency, and poor resource utilisation in agricultural supply chains.
Author's Approach/Method	Bibliographical research via Scopus and Web of Science (2018–2023). Keywords: "circular economy," "blockchain," "reverse logistics," "pesticide packaging." Used QR codes and Ethereum-based smart contracts written in Solidity.
Result	Explores how blockchain supports circular economy in agriculture through responsible disposal and recycling of pesticide packaging via reverse logistics [1].

Sl. No.	5
Title	Digital Twins for the Circular Economy — A. Preut, J.-P. Kopka, and U. Clausen (2021) [27]
Problem Addressed	Circular economy supply chains lack consistent lifecycle information and transparency; the digital twin concept lacks a unified definition.
Author's Approach/Method	Literature review, stakeholder analysis, and process analysis of circular supply chains. Develops a working definition of digital twin by comparing characteristics across multiple existing definitions [27].
Result	Digital twins can improve lifecycle information management, transparency, and coordination, enabling better repair, reuse, remanufacturing, and recycling decisions [27].

Sl. No.	6
Title	Blockchain Technology in the Food Supply Chain: A Way Towards Circular Economy and Sustainability — J. Zhang, Y. Liu, and H. Wang (2025) [17]
Problem Addressed	Food supply chains face inefficient resource management, food waste, and lack of traceability hindering circular economy implementation.
Author's Approach/Method	Proposes a blockchain-based traceability system integrating smart contracts to monitor supply chain transactions and resource flows from production to recycling.
Result	Blockchain improves traceability, transparency, and accountability, enabling better waste reduction strategies and promoting circular economy practices in supply chain management [17].

Sl. No.	7
Title	RockChain: A Blockchain-Driven Circular Waste Management Framework — A. Kumar, S. Patel, and R. Singh (2025) [18]
Problem Addressed	Construction and industrial sectors struggle to track waste materials and ensure their reintegration into circular supply chains.
Author's Approach/Method	Proposes the RockChain framework: a blockchain-based architecture recording material flows and waste transactions across stakeholders in a decentralised ledger.
Result	Demonstrates that blockchain can significantly improve traceability of waste materials, enhance recycling verification, and support circular resource management [18].

Sl. No.	8
Title	The Future is Traceable: Integrating Blockchain, Waste Management and Circular Economy — M. Rossi and L. Bianchi (2025) [19]
Problem Addressed	Current waste management systems lack reliable digital mechanisms to verify recycling processes and track the lifecycle of materials.
Author's Approach/Method	Proposes a blockchain-enabled waste tracking model incorporating digital product identities and decentralised data recording for lifecycle monitoring.
Result	Blockchain can improve trust, transparency, and verification of recycling activities, supporting accountable and measurable circular economy implementation [19].

Sl. No.	9
Title	A Systematic Literature Review on Blockchain for Circular Economy — E. Fernandez and R. Torres (2024) [20]
Problem Addressed	Despite growing interest, there is limited consolidated knowledge on existing architectures, applications, and research gaps.
Author's Approach/Method	Systematic literature review analysing recent academic publications on blockchain-enabled circular economy frameworks across multiple industries.
Result	Identifies key opportunities for blockchain adoption in lifecycle tracking, waste management, and sustainable supply chains; highlights challenges such as scalability, interoperability, and regulatory issues [20].

Sl. No.	10
Title	Optimizing Circular Economy Practices through Blockchain Technology in Reverse Logistics — S. Gupta, P. S. Kushwaha, and U. Badhera (2026) [15]
Problem Addressed	Very little work focuses on the circular economy within reverse logistics; traditional systems struggle with transparency, traceability, and trustworthiness.
Author's Approach/Method	Identified and validated critical circular economy behaviours using the Delphi technique and Grey Influential Analysis (GIA) to rank the most influential blockchain-enabled practices.
Result	A more holistic conceptual framework is necessary; the study provides a prioritised roadmap for organisations to focus on the most influential applications of blockchain in reverse logistics [15].

Sl. No.	11
Title	Blockchain-based Framework for Supply Chain Traceability: A Case Example of Textile and Clothing Industry — T. K. Agrawal et al. (2021) [23]
Problem Addressed	The textile industry suffers from illicit trades, unethical production practices, and opaque global supply chains leading to information asymmetry and counterfeiting.
Author's Approach/Method	Conceptualised supply chain architecture and smart contract rules, developing a prototype blockchain program to simulate a multi-tier organic cotton supply chain.
Result	Implemented a customised private blockchain using smart contracts to securely track organic cotton from fibre to retail, providing public transparency and immutable provenance [23].

Sl. No.	12
Title	Bringing Trust and Transparency to the Opaque World of Waste Management with Blockchain: A Polkadot Parathread Application — I. J. Scott, M. de Castro Neto, and F. L. Pinheiro (2023) [25]
Problem Addressed	Household waste management struggles with low consumer trust, unreliable information, and lack of incentive; existing blockchain proposals face scalability limitations and high fees.
Author's Approach/Method	Design Science research approach using the Substrate blockchain framework, with collaborative problem identification with municipalities.
Result	Created a hybrid blockchain solution integrating waste tracking, financial incentives, gamification, reduced transaction costs, and increased scalability to promote responsible recycling behaviours [25].

Sl. No.	13
Title	From Trash to Cash: How Blockchain and Multi-Sensor-Driven AI Can Transform Circular Economy of Plastic Waste — A. Chidepatil, P. Bindra, and D. Kulkarni (2021) [22]
Problem Addressed	Plastic manufacturers are reluctant to use recycled polymers due to lack of reliable quality information about recycled feedstock.
Author's Approach/Method	Integrated system combining AI-powered multi-sensor data fusion with a digital blockchain platform using smart contracts to facilitate transactions between segregators, recyclers, and manufacturers.
Result	System provides accurate data on recycled plastic quality and securely shares it across a trust-based ledger, incentivising manufacturers to purchase recycled feedstock rather than virgin materials [22].

Sl. No.	14
Title	Blockchain-Enabled Provenance Tracking for Sustainable Material Reuse in Construction Supply Chains — S. Wilson et al. (2024) [26]
Problem Addressed	Transition to circular economy in construction hindered by lack of reliable tracking for reused materials; high storage costs and privacy risks on public blockchains.
Author's Approach/Method	Multilayer architecture using customisable Polkadot parachains, IPFS for off-chain decentralised storage of Material Passports, and MongoDB for optimised data querying.
Result	High scalability and low gas costs; privacy through Role-Based Access Control; tamper-proof provenance trail for construction materials to definitively verify their suitability for reuse [26].

Sl. No.	15
Title	A Dynamic Trading Approach Based on Walrasian Equilibrium in a Blockchain-Based NFT Framework for Sustainable Waste Management — C. S. Kumar et al. (2025) [24]
Problem Addressed	Traditional waste management lacks transparency, risks user data privacy, and is vulnerable to unauthorised or fraudulent recycling organisations.
Author's Approach/Method	Created the Greenlink framework using zero-knowledge proofs to verify organisations without exposing sensitive data, executed on the Mina protocol and Polygon zkEVM.
Result	Dynamically balances supply and demand; securely verifies corporate compliance without exposing sensitive data; provides high-throughput, energy-efficient transaction processing [24].

Sl. No.	16
Title	Artificial Intelligence-Based Digital Twin Framework for Circular Economy Optimization in Healthcare Waste Management — B. Biswas and J. O. Akomodi (2026) [9]
Problem Addressed	Healthcare systems produce increasing waste; traditional linear waste management is inefficient and fails to comply with circular economy principles.
Author's Approach/Method	Proposes a digital twin system virtualising hospital waste streams via IoT-based monitoring, integrating AI and ML to forecast waste patterns, with a multi-objective optimisation engine.
Result	Average recycling rate raised from 20.0% to 50.7% (a 30.7 percentage-point increase), verified by sample t-testing at $p < 0.001$ with statistical power of 0.99 [9].

Sl. No.	17
Title	Blockchain-Digital Twin Integration in Construction Material Passports: Preparing for Digital Product Passport Implementation — S. Aklashie et al. (2026) [12]
Problem Addressed	Construction industry transition to circular economy is hindered by poor material lifecycle data management; existing Material Passport frameworks operate in isolation.
Author's Approach/Method	Systematic literature review following PRISMA framework: bibliometric analysis of 62 publications and content analysis of 29 studies sourced from Scopus and Web of Science.
Result	Integrating blockchain and digital twins creates a synergistic effect; barriers identified include technical interoperability issues, data storage conflicts, and lack of standardisation [12].

Sl. No.	18
Title	Leveraging Blockchain and Digital Twins for Low-Carbon, Circular Supply Chains: Evidence from the Moroccan Manufacturing Sector — S. Abdallah-Ou-Moussa, M. Wynn, and Z. Rouaine (2026) [14]
Problem Addressed	Global supply chains struggle to reconcile economic efficiency with environmental responsibility; existing research on blockchain and digital twins is often fragmented.
Author's Approach/Method	Explanatory sequential mixed-methods design: integrative literature review, 30 semi-structured interviews, and a quantitative survey of 125 Moroccan manufacturing firms analysed using PLS-SEM.
Result	Blockchain improves traceability and governance; digital twins enhance eco-efficiency. Circular Integration acts as an "effect multiplier," explaining 68% of variance in environmental performance ($R^2 = 0.68$) [14].

Sl. No.	19
Title	The Role of Digital Twins in Shaping the Circular Economy for SMEs in Emerging Markets — M. M. Shoily (2026) [16]
Problem Addressed	SMEs in emerging economies face technological, financial, and information barriers when transitioning to a circular economy.
Author's Approach/Method	Study of available literature and emerging case studies describing how digital twins can be utilised in resource-poor settings for lifecycle tracking and predictive maintenance.
Result	Digital Twin technology has "tremendous potential" for SMEs, contingent on overcoming high initial costs and gaps in data management and technical know-how [16].

Sl. No.	20
Title	Knowledge-Based Optimization Algorithm for Feature Selection in Promoting Circular Economy through Blockchain — S. Sridevi, B. V. Kumar, and G. R. Karpagam (2026) [13]
Problem Addressed	Blockchain integrity is threatened by malicious miners (selfish mining, block withholding); conventional optimisation algorithms suffer from slow convergence and suboptimal results.
Author's Approach/Method	Developed Knowledge-based GA (KBGA) and Knowledge-based DE (KBDE) using Knowledge-based Initialisation (k-means clustering) and Knowledge-based Selective Mutation. 213 features extracted across four blockchain layers; Random Forest classifier used for miner loyalty prediction.
Result	KBDE-RF achieved 8.9% higher average accuracy than standard GA-RF and reached optimal solutions in far fewer generations, also reducing power consumption compared to Proof-of-Work systems [13].

Analysis of Literature Survey

The twenty papers reviewed in this survey span eight thematic areas that are directly relevant to the proposed blockchain-enabled Digital Product Passport (DPP) system for circular economy implementation. The analysis below examines each theme in turn, identifying key contributions and the limitations that motivate the proposed system.

Theme 1: Blockchain Technology as Core Infrastructure. Seventeen of the twenty surveyed papers address blockchain as their primary or supporting technology, confirming it as the dominant enabler for transparent and tamper-proof data management across industries. Böckel et al. [4] and Fernandez and Torres [20] provide foundational overviews, documenting the research-to-practice gap and identifying that most real-world deployments have not yet achieved the trust and interoperability promised by academic frameworks. Yli-Huumo et al. [6] and Dong et al. [7] survey the broader blockchain landscape, establishing that immutability, transparency, and decentralised consensus are the core properties that supply chain and lifecycle systems require. Chang et al. [8] and Hübschke et al. [10] further examine implementation

success factors and measurement methods, noting that scalability, high transaction costs on public networks, and lack of enterprise integration remain persistent barriers. Karaduman and Gülhas [11] specifically highlight security and data integrity as unresolved challenges. Across these works, the consensus is that while blockchain provides a strong technical foundation, the validation inputs it receives — primarily QR scans and manual entries — are inherently weak, undermining the integrity of the entire system.

Theme 2: Circular Economy Principles and Frameworks. Fourteen papers directly engage with circular economy (CE) principles, making it the second most prevalent theme. Böckel et al. [4] analyse the gap between CE theory and practice, finding that the lack of reliable verification mechanisms is the primary barrier to adoption. Kumar and Chopra [3] propose a Circular Economy Blockchain (CEB) architecture that addresses coordination and trust deficits. Fernandez and Torres [20] conduct a systematic review identifying scalability, interoperability, and regulatory compliance as the top challenges for blockchain-enabled CE systems. Nandi et al. [21] highlight how the COVID-19 pandemic exposed the fragility of linear supply chains and accelerated interest in CE-based redesign, providing a strong contemporary motivation for this work. Prause et al. [1], Zhang et al. [17], Gupta et al. [15], and Sridevi et al. [13] extend CE analysis to agriculture, food supply chains, reverse logistics, and blockchain consensus mechanisms respectively, each demonstrating the breadth of CE applicability while reinforcing the common finding that data authenticity is the key unsolved problem.

Theme 3: Digital Product Passports and Lifecycle Tracking. Only three papers explicitly address Digital Product Passports (DPPs) — Xia et al. [5], Wilson et al. [26], and Aklashie et al. [12] — making this the least-studied theme despite being the central mechanism of the proposed system. Xia et al. [5] provide the most directly relevant work, demonstrating how blockchain-enabled DPPs can improve traceability and information sharing in reverse supply chains and highlighting the limitations of QR-scan-based lifecycle updates. Wilson et al. [26] implement a multilayer blockchain architecture for material provenance tracking in construction, using IPFS for off-chain DPP storage and Polkadot parachains for privacy-preserving access control. Aklashie et al. [12] prepare a theoretical foundation for DPP implementation by reviewing the integration of blockchain and digital twins in construction material passports, identifying interoperability and real-time synchronisation as unresolved challenges. The scarcity of DPP-focused research, despite the EU Digital Product Passport regulation coming into effect, confirms that this is an emerging and critically important area and directly motivates the proposed system's DPP-centric design.

Theme 4: Supply Chain Traceability and Transparency. Eight papers address supply chain traceability, spanning agriculture [1], food [17], textiles [23], construction [26], and general circular supply chains [5, 10, 18]. Agrawal et al. [23] provide a concrete case study of a private blockchain tracking organic cotton from fibre to retail, achieving immutable provenance and demonstrating the feasibility of multi-tier supply chain implementation. Xia et al. [5] specifically address reverse supply chains, showing that DPPs improve both upstream and downstream traceability. Abdallah-Ou-Moussa et al. [14] quantitatively demonstrate that blockchain adoption, combined with digital twins, explains 68% of the variance in environmental performance ($R^2 = 0.68$) in manufacturing supply chains. Preut et al. [27] and Gupta et al. [15] address circular supply chains specifically, finding that stakeholder coordination and lack of standardised data formats are the primary obstacles. A consistent limitation across this theme is that supply chain events are validated through user-triggered scans rather than verified economic activity, creating opportunities for data manipulation at each handoff point.

Theme 5: Waste Management and Recycling. Seven papers focus specifically on waste management systems, with plastic waste and construction debris being the primary domains. Scott et al. [25] implement a Polkadot parathread application for household waste management, incorporating financial incentives and gamification to drive consumer participation — a direct precursor to the token-based incentive model in the proposed system. Chidepatil et al. [22] combine AI-driven sensor data with blockchain to verify the quality of recycled plastic feedstock, solving the manufacturer reluctance problem at the recycling end of the chain. Kumar et al. [18] introduce RockChain for construction waste, demonstrating decentralised ledger tracking of material flows. Rossi and Bianchi [19] argue that blockchain is the missing verification layer in waste management, while Biswas and Akomodi [9] show in a healthcare context that AI-integrated digital twin frameworks can raise recycling rates from 20% to 50.7%. A key finding across these works is that verified recycling events — rather than self-reported or scan-confirmed ones — are essential for credible sustainability claims, reinforcing the proposed system's reliance on payment-linked validation.

Theme 6: Smart Contracts and Payment-Linked Validation. Smart contracts are addressed in four papers: Kumar and Chopra [3], Zhang et al. [17], Agrawal et al. [23], and Kumar et al. [24]. Kumar and Chopra [3] establish the theoretical case for smart-contract-driven CE implementation, proposing automatic execution of lifecycle events once predefined conditions are met. Zhang et al. [17] implement smart contracts to monitor food supply chain resource flows, confirming their feasibility for production-to-recycling event recording. Agrawal et al. [23] demonstrate smart contract validation of multi-tier organic cotton provenance. Kumar et al. [24] apply a

Walrasian equilibrium model within a blockchain-NFT framework for waste trading, showing that economically-grounded transactions provide stronger verification signals than user-triggered scans. Critically, none of the four papers links smart contract execution directly to financial payment events as the primary proof of real-world activity. This represents the defining research gap addressed by the proposed system, in which UPI or token-based payment transactions serve as the trigger and proof of lifecycle events such as product return, collection, and recycling.

Theme 7: Digital Twins and IoT Integration. Six papers address digital twins as a complementary or alternative technology: Preut et al. [27], Biswas and Akomodi [9], Aklashie et al. [12], Abdallah-Ou-Moussa et al. [14], and Shoily [16]. Preut et al. [27] provide the foundational definition and stakeholder analysis, concluding that digital twins offer significant potential for lifecycle information management in circular supply chains but lack the cryptographic security needed for tamper-proof recording. Aklashie et al. [12] directly address this gap, showing that the combination of blockchain (for immutability) and digital twins (for real-time state monitoring) creates a synergistic effect superior to either technology alone. Biswas and Akomodi [9] demonstrate an AI-digital twin framework achieving a 30.7 percentage point improvement in healthcare waste recycling rates. Abdallah-Ou-Moussa et al. [14] and Shoily [16] highlight that the primary barriers to digital twin adoption — particularly in SMEs and emerging markets — are high initial costs and insufficient data management infrastructure. These findings position digital twin integration as a valuable future extension of the proposed system's IoT layer, while blockchain remains the more practical near-term foundation.

Theme 8: Artificial Intelligence and Optimisation. Three papers integrate AI and machine learning techniques. Chidepatil et al. [22] combine multi-sensor AI fusion with blockchain to assess recycled plastic quality, addressing manufacturer reluctance by providing verifiable feedstock data. Biswas and Akomodi [9] use ML-based waste forecasting within a digital twin framework, achieving statistically significant improvements in recycling efficiency ($p < 0.001$). Sridevi et al. [13] apply knowledge-based genetic and differential evolution algorithms (KBGA and KBDE) to the problem of blockchain miner loyalty prediction, reducing malicious mining activity and improving system reliability. While AI is not a primary component of the current proposed system, these papers collectively demonstrate that AI-driven analytics over verified blockchain data can substantially improve circular economy outcomes, and establish a pathway for incorporating predictive capabilities in future iterations.

Identified Research Gap and Positioning of the Proposed System. The thematic analysis reveals a clear and consistent gap: no existing work combines all five of the core components required for a trustworthy, end-to-end circular economy system — namely, (1) blockchain-based immutable recording, (2) Digital Product Passports for standardised lifecycle identity, (3) payment-linked validation as proof of real-world events, (4) token-based consumer incentives for recycling participation, and (5) automated ESG reporting from verified transaction data. While individual papers address two or three of these components in isolation, the absence of an integrated framework that ties financial transactions to lifecycle state changes is the fundamental limitation. Most existing systems continue to rely on QR scans, manual data entry, or user-reported updates — mechanisms that Böckel et al. [4], Kumar and Chopra [3], and Fernandez and Torres [20] collectively identify as the primary source of unreliable sustainability claims. The proposed DPP system directly addresses this gap by replacing scan-based verification with payment-linked validation, ensuring that every lifecycle update recorded on the blockchain corresponds to a verifiable economic transaction, thereby establishing a stronger and more fraud-resistant foundation for circular economy implementation.

SYSTEM REQUIREMENT SPECIFICATION

5.1 Functional Requirements

FR-1: DPP Creation

- Manufacturer creates DPP
- Assigns DID (Decentralized ID)
- Generates QR (identity link only)

FR-2: Authentication (QR Role)

- QR scan retrieves DPP
- No state change occurs

FR-3: Payment-linked Lifecycle Events

- Ownership transfer → payment settlement
- Recycling → incentive payment
- Smart contract validates transaction

FR-4: Consumer Incentives

- User receives tokens/UPI cashback
- Linked to verified recycling event

FR-5: Recycler Processing

- Recycler logs processing
- Payment to system triggers DPP closure

FR-6: Reporting

- Auto-generated ESG reports
- Based on verified financial transactions

5.2 Non-functional Requirements

- **Security:** Cryptographic validation

- **Performance:** <5 second transaction time
- **Privacy:** No personally identifiable information (PII) stored
- **Scalability:** Microservices and containers
- **Reliability:** Offline sync support

5.3 Hardware Requirements

- Mobile devices (QR scan)
- Servers / cloud infrastructure
- Optional IoT sensors

5.4 Software Requirements

- **Blockchain:** Hyperledger Fabric / Ethereum
- **Payment integration:** UPI / token system
- **Backend:** REST APIs
- **Storage:** IPFS

SYSTEM DESIGN

6.1 System Design

6.1.1 System Architecture

Layers:

1. **User Layer**
    - Manufacturer
    - Consumer
    - Recycler
2. **Authentication Layer** — QR → fetch DPP
3. **Payment Layer**
    - UPI / token transactions
    - Wallet integration
4. **Blockchain Layer**
    - Smart contracts validate payments
    - Update lifecycle
5. **Storage Layer**
    - On-chain (hashes)
    - Off-chain (IPFS)

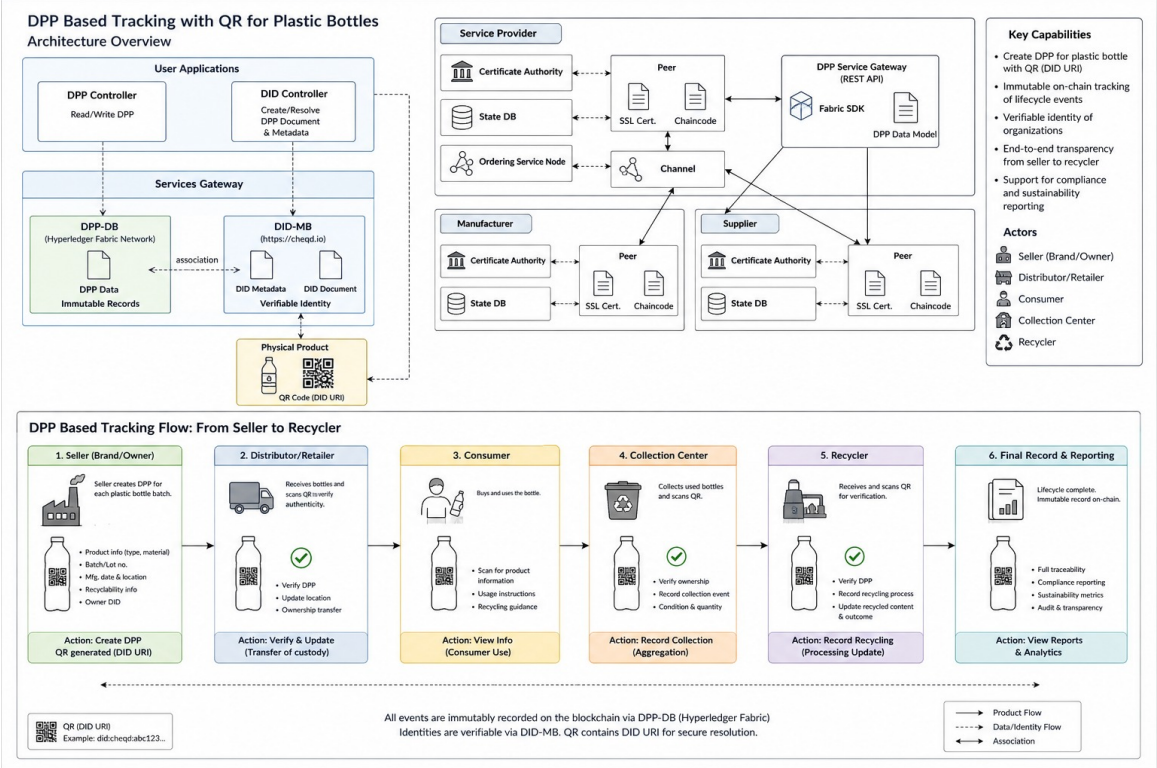


Figure 6.1: System Architecture – Blockchain-based Digital Product Passport

As illustrated in Figure 6.1, the architecture represents a blockchain-based Digital Product Passport system designed to enable transparent and secure lifecycle tracking of plastic bottles within a circular economy [5, 7]. The system is built on Hyperledger Fabric and is structured into three main layers: user applications, service gateway, and blockchain network. The user application layer includes the DPP Controller, which manages lifecycle data, and the DID Controller, which handles decentralized identity creation and verification. These components interact with the service gateway, which connects to the DPP database storing immutable lifecycle records and the DID metadata system that ensures verifiable identities. Each physical product is linked to a QR code containing a DID URI, which is used only for authentication and retrieving product information, not for updating system states [5].

The blockchain layer consists of peers, certificate authorities, ordering services, channels, and chaincode. Peers maintain the distributed ledger, certificate authorities issue secure identities, and the ordering service ensures consensus across the network [7]. Chaincode defines the business logic and governs how transactions are validated and recorded. Stakeholders such as manufacturers, distributors, and recyclers operate nodes within this network, ensuring decentralized control and transparency [6]. The architecture enables capabilities such as immutable lifecycle tracking, identity verification, and regulatory compliance, while maintaining scalability and security.

The lifecycle flow begins with the manufacturer creating a Digital Product Passport and generating a QR code linked to the product. Distributors verify and update ownership through blockchain transactions [5]. Consumers can scan the QR code to access product information without altering its state. When the product reaches a collection center, it is aggregated and prepared for recycling. The recycler processes the material and updates the lifecycle status on the blockchain. In an enhanced model, lifecycle updates are linked to payment transactions, ensuring that events such as recycling are validated through real economic activity rather than simple scans [5]. This approach improves trust, reduces fraud, and ensures accurate representation of physical processes.

Additionally, the system supports automated reporting and analytics by aggregating verified lifecycle data into compliance-ready formats, enabling organizations to meet regulatory requirements and sustainability goals [14]. The integration of payment-linked validation further strengthens accountability by ensuring that every recorded event corresponds to a confirmed transaction, thereby bridging the gap between digital records and real-world actions. This makes the system not only technologically robust but also practically viable for large-scale adoption.

6.1.2 Module Design

High Level Design

Figure 6.2 presents the high-level design of the system, illustrating the key components and their interactions across the DPP lifecycle.

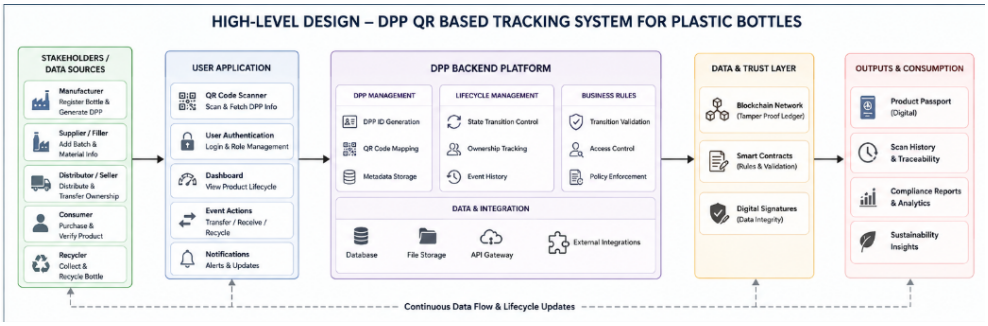


Figure 6.2: High Level Design

Low Level Design

Figure 6.3 provides the low-level design, detailing the internal module structure and data flow within each architectural layer.

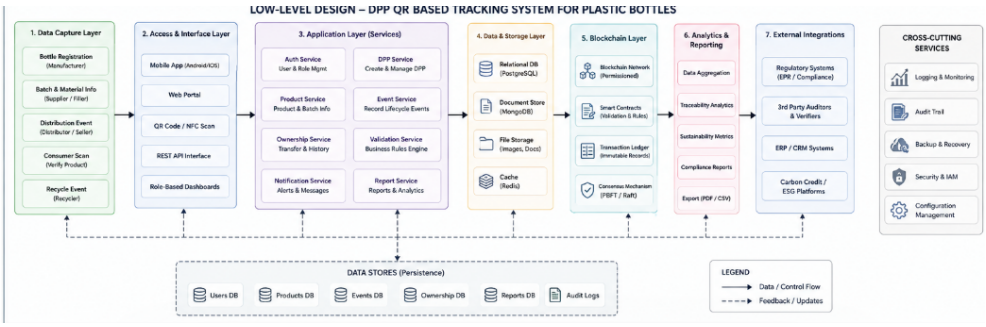


Figure 6.3: Low Level Design

6.2 Detailed Design

6.2.1 Class Diagram

Figure 6.4 shows the class diagram of the system, depicting the primary entities—DPP, DID, QR Code, Payment, and Stakeholder—along with their attributes and associations.

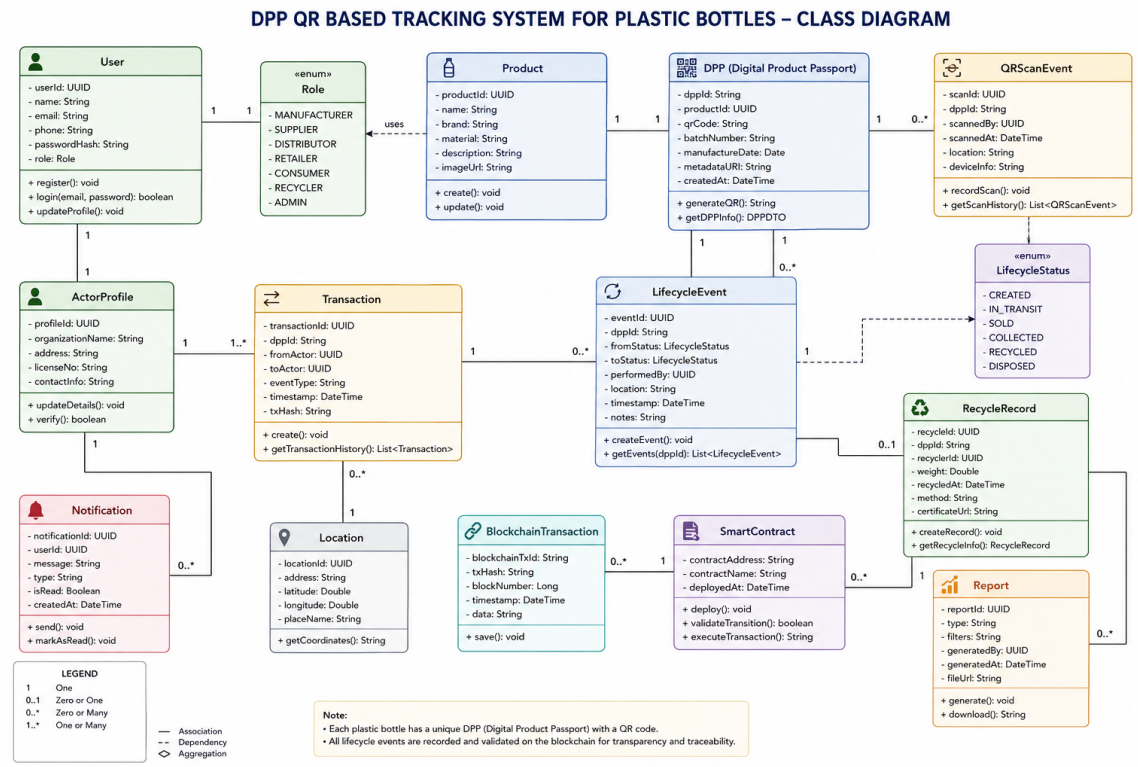


Figure 6.4: Class Diagram

6.2.2 Activity Diagram

The activity diagram in Figure 6.5 illustrates the sequential flow of operations from DPP creation through payment-linked lifecycle events to final recycling and reporting.

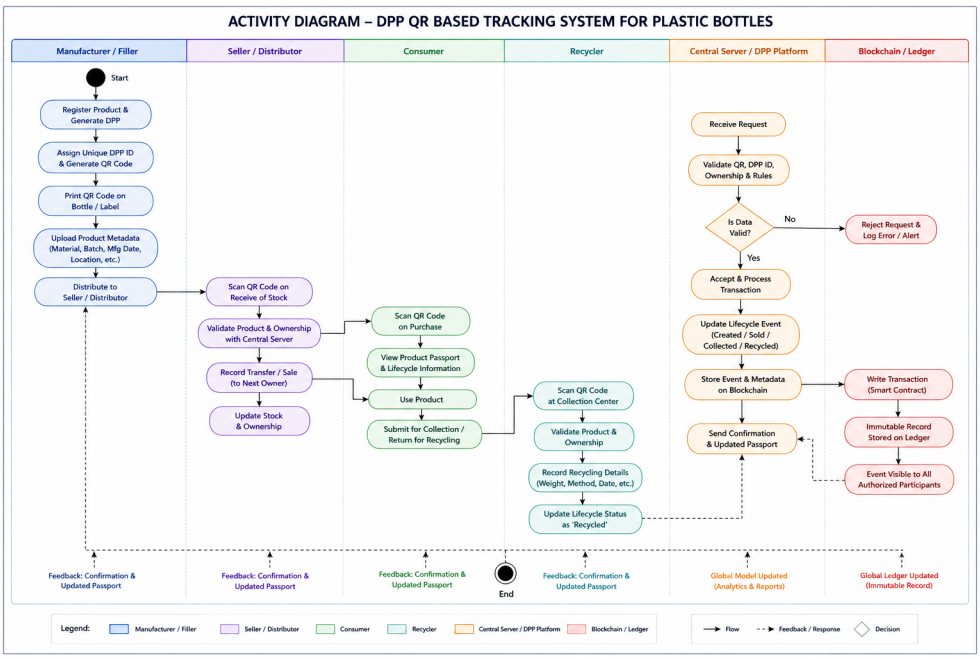


Figure 6.5: Activity Diagram

6.2.3 Use Case Diagram

Figure 6.6 presents the use case diagram, capturing the interactions between the three primary actors—Manufacturer, Consumer, and Recycler—and the core system functions.

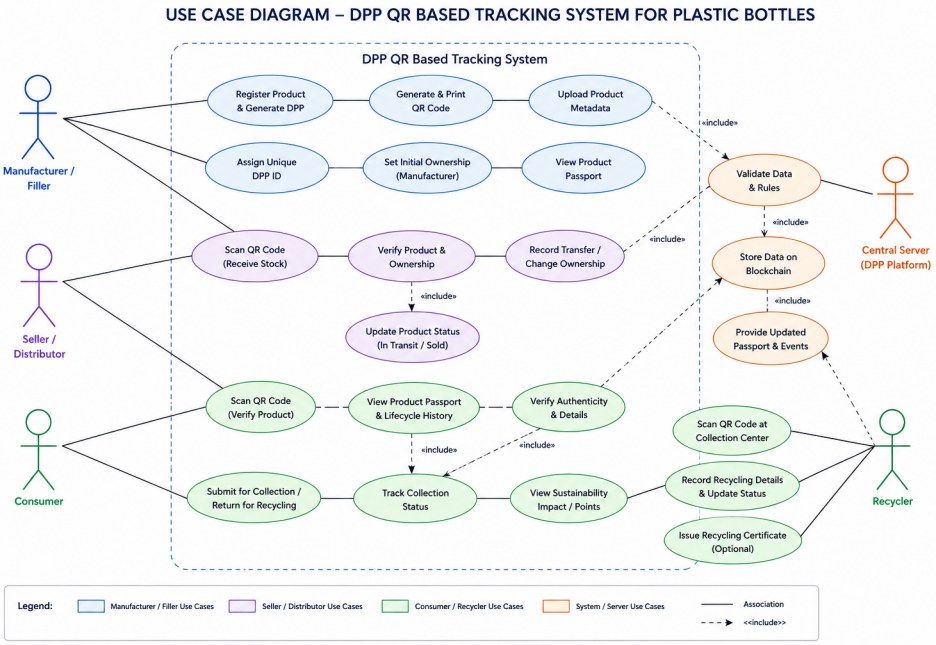


Figure 6.6: Use Case Diagram

6.2.4 Scenarios

Scenario 1: End-to-End Supply Chain Transparency

Situation: Multiple stakeholders such as manufacturers, distributors, retailers, and recyclers are involved in the lifecycle of plastic bottles. However, there is a lack of transparency in tracking product movement and ownership across the supply chain [8].

System Working: Each plastic bottle is assigned a unique DPP ID and QR code at the manufacturing stage, as shown in Figure 6.1. As the bottle moves through the supply chain, each stakeholder scans the QR code to record ownership transfer and update product status. These events are validated and stored on the blockchain through the central DPP platform [5].

Outcome: A complete and tamper-proof history of the product lifecycle is maintained [6]. All stakeholders can verify product origin, movement, and authenticity, ensuring transparency and accountability across the supply chain.

Scenario 2: Fraudulent / Duplicate Product Detection

Situation: Counterfeit or duplicate plastic products enter the supply chain, leading to financial losses and lack of trust among stakeholders [11].

System Working: When a QR code is scanned, the system checks the DPP ID against the blockchain ledger, as described in the activity flow of Figure 6.5. If the same QR code is scanned multiple times in conflicting locations or ownership chains, the system flags it as suspicious. Invalid or duplicate transactions are rejected by the validation module [5].

Outcome: Fraudulent or duplicate products are quickly detected and prevented from circulating further [6]. This enhances trust, ensures product authenticity, and protects both businesses and consumers.

Scenario 3: Improper Disposal / Recycling Non-Compliance

Situation: Consumers often dispose of plastic bottles improperly, and recyclers may fail to report accurate recycling data, leading to poor waste management [22].

System Working: Consumers scan the QR code while returning bottles to authorized collection centers (see Figure 6.6). The recycler scans and records recycling details such as method, weight, and timestamp. The system verifies the legitimacy of the recycling event before updating the lifecycle status to “Recycled” [5].

Outcome: Only verified recycling activities are recorded, ensuring accurate waste

tracking [25]. This promotes responsible disposal behavior and improves compliance with environmental regulations.

Scenario 4: Consumer Awareness & Sustainability Insights

Situation: Consumers lack visibility into the environmental impact and lifecycle of the plastic products they use.

System Working: By scanning the QR code, consumers can access the Digital Product Passport, which includes details such as material composition, origin, carbon footprint, and recycling instructions [5]. The system also tracks user participation in recycling activities, as illustrated in Figure 6.6.

Outcome: Consumers become more informed and environmentally conscious [22]. The system encourages sustainable behavior and enables data-driven insights into recycling patterns and environmental impact.

APPLICATIONS

1. Supply Chain Transparency

The system enables complete visibility across the plastic bottle supply chain, from manufacturing to recycling [5, 8]. As depicted in Figure 6.1, each bottle is assigned a unique QR-linked Digital Product Passport, allowing stakeholders to track ownership and movement in real time. This ensures accountability at every stage, reduces data inconsistencies, and builds trust among manufacturers, distributors, and regulators [6]. The immutable blockchain records prevent tampering, ensuring that all transactions and lifecycle updates remain transparent and verifiable.

2. Anti-Counterfeiting and Product Authentication

By using unique QR codes linked to secure DPP records, the system helps detect counterfeit or duplicate products in the market [11]. Each scan validates the product's authenticity against blockchain-stored data. If inconsistencies arise, such as duplicate scans or mismatched ownership, the system flags them immediately [5, 10]. This protects brands from fraud, ensures genuine product distribution, and provides consumers with confidence in the authenticity and safety of the plastic products they purchase.

3. Efficient Waste Management and Recycling

The system improves waste management by tracking plastic bottles through their entire lifecycle, including disposal and recycling [22, 25]. Collection centers and recyclers update recycling data by scanning QR codes, ensuring accurate recording of waste processing activities [18, 19]. This enables better monitoring of recycling rates and reduces unaccounted waste. Governments and organizations can use this data to optimize waste collection systems and promote environmentally responsible disposal practices among consumers.

4. Regulatory Compliance and Reporting

DPP systems support compliance with environmental regulations by maintaining detailed, verifiable records of product lifecycle events [1, 3]. Authorities can access real-time data on production, distribution, and recycling activities. This simplifies auditing processes and ensures adherence to policies such as Extended Producer Responsibility (EPR). Automated reporting features generate compliance reports, reducing manual effort and improving accuracy, while helping organizations meet sustainability and legal requirements effectively [14].

5. Consumer Engagement and Sustainability Awareness

The system enhances consumer awareness by providing easy access to product information through QR scans [5]. Users can view details such as material composition, origin, and recycling instructions. Additionally, the platform can reward users for responsible disposal actions, encouraging participation in recycling programs [22, 25]. This fosters environmentally conscious behavior and strengthens the connection between consumers and sustainable practices, ultimately contributing to reduced plastic waste and improved environmental outcomes.

CONCLUSIONS

The implementation of a Digital Product Passport (DPP) integrated with QR-based tracking for plastic waste management presents a transformative approach to addressing the growing global plastic pollution crisis [5]. This system enhances transparency across the entire lifecycle of plastic products by enabling real-time traceability from production to disposal and recycling [5, 8]. By assigning unique QR codes to plastic items, stakeholders—including manufacturers, consumers, waste collectors, and recycling facilities—can access detailed information about material composition, origin, usage history, and recommended end-of-life treatment.

The proposed system significantly improves accountability within the plastic value chain [3]. Manufacturers are encouraged to adopt sustainable materials and eco-friendly packaging due to increased visibility of product lifecycle impacts. Consumers, in turn, become more informed about disposal practices, leading to higher participation in segregation and recycling activities [22]. Additionally, waste management authorities benefit from structured data collection, which supports more efficient sorting, reduced contamination in recycling streams, and improved resource recovery rates [19, 18].

From a technological perspective, integrating blockchain or secure cloud-based databases with QR-enabled DPP systems ensures data integrity and prevents tampering, thereby strengthening trust among stakeholders [6, 7]. However, challenges such as initial implementation costs, standardization across industries, and user adoption barriers must be addressed for large-scale deployment [20, 10].

Overall, the DPP QR-based tracking system represents a scalable and impactful solution aligned with circular economy principles [21]. It bridges critical gaps in current waste management systems by enabling data-driven decision-making and fostering sustainable consumption patterns [5, 4]. With continued technological advancement and policy support, this approach has the potential to significantly reduce plastic waste leakage into the environment and promote a more sustainable and responsible materials ecosystem.

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